

Lista 2 de Exercícios de álgebra linear

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Problema 1. Determine o núcleo e a imagem das transformações lineares dadas pelas matrizes a seguir (neste problema, estamos sempre trabalhando com $\mathbb{R}^n$ e a base canonica).

(a) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix}.$

(b) $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}.$

(c) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$

(d) $\begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$

Problema 2. Determine quais das matrizes abaixo são inversíveis, e para as que forem, diga a sua inversa.

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & -24 \end{bmatrix}.$

(b) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$

(c) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & -9 \end{bmatrix}.$

Problema 3. Calcule o determinante das matrizes do problema 2.

• Problema 1:

Problema 1. Determine o núcleo e a imagem das transformações lineares dadas pelas matrizes a seguir (neste problema, estamos sempre trabalhando com $\mathbb{R}^n$ e a base canônica).

ESTRATÉGIA: ESCALONAR T/ IDENTIFICAR SE HÁ VETOR COLUNA $\mathbf{h}_0$.

$$(a) \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \xrightarrow{\text{ESCALONAR}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

a.1) NÚCLEO

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \leadsto \begin{matrix} x + 0 - z = 0 & x = z \\ 0 + y + z = 0 & y = -z \\ 0 + 0 + 0 = 0 \end{matrix} \therefore x = -y = z \therefore t(1, -1, 1)$$

$$\text{Logo: } \text{Ker}(T) = \langle (1, -1, 1) \rangle$$

a.2) Imagem

$$\text{Logo: } \text{Im}(T) = \langle (1, 0, -1), (1, 1, 0) \rangle$$

$$(b) \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \xrightarrow{h_3 = h_3 + h_1} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{h_2 = h_2 + h_3} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

a.1) NÚCLEO

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \leadsto \begin{matrix} x - y + 0 = 0 \\ 0 + 0 + 0 = 0 \\ 0 - y + z = 0 \end{matrix} \therefore x = y = z$$

$$\text{Logo: } \text{Ker}(T) = \langle (1, 1, 1) \rangle$$

a.2) Imagem

$$\text{Logo: } \text{Im}(T) = \langle (1, 0, -1), (-1, 1, 0) \rangle$$

$$(c) \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{\text{ESCALONANDO}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

a.1) NÚCLEO

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \leadsto \begin{matrix} x + 0 - z = 0 & x = z \\ 0 + y + z = 0 & y = -z \\ 0 + 0 + 0 = 0 \end{matrix} \therefore t(1, -1, 1)$$

$$\text{Logo: } \text{Ker}(T) = \langle (1, -1, 1) \rangle$$

a.2) Imagem

$$\text{Logo: } \text{Im}(T) = \langle (1, 0, 1), (1, 1, 0) \rangle$$

$$(d) \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{h_3 = h_3 + h_1 \\ h_2 = h_2 + h_1}} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{h_1 = h_1 + h_3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{h_2 = h_2 - h_3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{h_3 = h_3 + h_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

a.1) NÚCLEO

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \leadsto \begin{matrix} x = z \\ y = z \end{matrix} \therefore t(1, 1, 1)$$

$$\text{Logo: } \text{Ker}(T) = \langle (-1, 1, 1) \rangle$$

a.2) Imagem

$$\text{Logo: } \text{Im}(T) = \langle (1, -1, -1), (-1, 1, 0) \rangle$$

Problema 2. Determine quais das matrizes abaixo são inversíveis, e para as que forem, diga a sua inversa.

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & -24 \end{bmatrix}$.

a.1) $\det(A) = 8$

$$\left[\begin{array}{ccc|cc} 1 & 2 & 3 & 1 & 2 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 1 & -24 & 3 & 1 \end{array} \right] \therefore [(1 \cdot 3 \cdot -24) + (2 \cdot 1 \cdot 3) + (3 \cdot 2 \cdot 1)] - [(3 \cdot 3 \cdot 3) + (1 \cdot 1 \cdot 1) + (24 \cdot 2 \cdot 2)] =$$

$$[-72 + 6 + 6] - [9 + 1 + 96] = [-60] - [106] = -166$$

logo, como $\det \neq 0$, matriz é inversível.

a.2) Inversa

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 3 & 1 & -24 & 0 & 0 & 1 \end{array} \right] \xrightarrow{h_2 = h_2 - 2h_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -5 & -2 & 1 & 0 \\ 0 & -5 & -33 & -3 & 0 & 1 \end{array} \right] \xrightarrow{h_3 = h_3 - 5h_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -5 & -2 & 1 & 0 \\ 0 & 0 & 1 & 7 & 5 & 1 \end{array} \right]$$

$$\xrightarrow{h_1 = h_1 + 2h_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -7 & -3 & 2 & 0 \\ 0 & -1 & -5 & -2 & 1 & 0 \\ 0 & 0 & 1 & 7 & 5 & 1 \end{array} \right] \xrightarrow{h_2 = -h_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -7 & -3 & 2 & 0 \\ 0 & 1 & 5 & 2 & -1 & 0 \\ 0 & 0 & 1 & 7 & 5 & 1 \end{array} \right] \xrightarrow{h_1 = h_1 + 7h_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -54 & -13 & -7 \\ 0 & 1 & 0 & -12 & -6 & -5 \\ 0 & 0 & 1 & 7 & 5 & 1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} -54 & -13 & -7 \\ -12 & -6 & -5 \\ 7 & 5 & 1 \end{bmatrix}$$

(b) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$.

b.1) $\det(B) = -1$

$$\left[\begin{array}{ccc|cc} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & -2 & 1 & 0 \end{array} \right] \therefore [(1 \cdot 1 \cdot -2) + (1 \cdot 1 \cdot 1) + (0 \cdot 0 \cdot 0)] - [(1 \cdot 1 \cdot 0) + (0 \cdot 1 \cdot 1) + (-2 \cdot 0 \cdot 1)] =$$

$$[-2 + 1 + 0] - [0 + 0 + 0] = -1$$

logo, como $\det \neq 0$, matriz é inversível.

b.2)

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & -2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{h_3 = h_3 - h_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & -1 & 0 & 1 \end{array} \right] \xrightarrow{h_1 = h_1 - h_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & -1 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{h_3 = h_3 + h_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 & 1 \end{array} \right] \xrightarrow{h_1 = h_1 + h_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -2 & -1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 & 1 \end{array} \right] \xrightarrow{h_2 = h_2 - h_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -2 & -1 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & -1 & -1 & 1 \end{array} \right]$$

$$B^{-1} = \begin{bmatrix} 0 & -2 & -1 \\ 1 & 2 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$

(c) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & -9 \end{bmatrix}$.

c.1) $\det(B) = -1$

$$\left[\begin{array}{ccc|cc} 1 & 2 & 0 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 \\ 2 & 0 & -9 & 2 & 0 \end{array} \right] \therefore [(1 \cdot 1 \cdot -9) + (2 \cdot 2 \cdot 2) + (0 \cdot 0 \cdot 0)] - [(2 \cdot 1 \cdot 0) + (0 \cdot 2 \cdot 1) + (-9 \cdot 0 \cdot 2)] =$$

$$[-9 + 8 + 0] - [0 + 0 + 0] = -1$$

logo, como $\det \neq 0$, matriz é inversível.

c.2)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 2 & 0 & -9 & 0 & 0 & 1 \end{array} \right] \xrightarrow{h_3 = h_3 - 2h_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & -4 & -9 & -2 & 0 & 1 \end{array} \right] \xrightarrow{h_1 = h_1 - 2h_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -4 & 1 & -2 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & -4 & -9 & -2 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{h_3 = h_3 + 4h_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -4 & 1 & -2 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 & 4 & 1 \end{array} \right] \xrightarrow{h_1 = h_1 + 4h_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -7 & 18 & 4 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 & 4 & 1 \end{array} \right] \xrightarrow{h_2 = h_2 - 2h_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -7 & 18 & 4 \\ 0 & 1 & 0 & 4 & 9 & 2 \\ 0 & 0 & -1 & -2 & 4 & 1 \end{array} \right]$$

$$C^{-1} = \begin{bmatrix} -7 & 18 & 4 \\ 4 & 9 & 2 \\ -2 & 4 & 1 \end{bmatrix}$$

Problema 3. Calcule o determinante das matrizes do problema 2.

a.) $\det(A) = 8$

$$\left[\begin{array}{ccc|cc} 1 & 2 & 3 & 1 & 2 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 1 & -24 & 3 & 1 \end{array} \right] \therefore [(1 \cdot 3 \cdot -24) + (2 \cdot 1 \cdot 3) + (3 \cdot 2 \cdot 1)] - [(3 \cdot 3 \cdot 3) + (1 \cdot 1 \cdot 1) + (24 \cdot 2 \cdot 2)] =$$
$$[-72 + 6 + 6] - [9 + 1 + 96] = [-60] - [106] = -166$$

Logo, como $\det \neq 0$, matriz é invertível.

b.) $\det(B) = -1$

$$\left[\begin{array}{ccc|cc} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & -2 & 1 & 0 \end{array} \right] \therefore [(1 \cdot 1 \cdot -2) + (1 \cdot 1 \cdot 1) + (0 \cdot 0 \cdot 0)] - [(1 \cdot 1 \cdot 0) + (0 \cdot 1 \cdot 1) + (-2 \cdot 0 \cdot 1)] =$$
$$[-2 + 1 + 0] - [0 + 0 + 0] = -1$$

Logo, como $\det \neq 0$, matriz é invertível.

c.) $\det(B) = -1$

$$\left[\begin{array}{ccc|cc} 1 & 2 & 0 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 \\ 2 & 0 & -9 & 2 & 0 \end{array} \right] \therefore [(1 \cdot 1 \cdot -9) + (2 \cdot 2 \cdot 2) + (0 \cdot 0 \cdot 0)] - [(2 \cdot 1 \cdot 0) + (0 \cdot 2 \cdot 1) + (-9 \cdot 0 \cdot 2)] =$$
$$[-9 + 8 + 0] - [0 + 0 + 0] = -1$$

Logo, como $\det \neq 0$, matriz é invertível.